

3.2 Management of threats to the Antarctic ecosystem

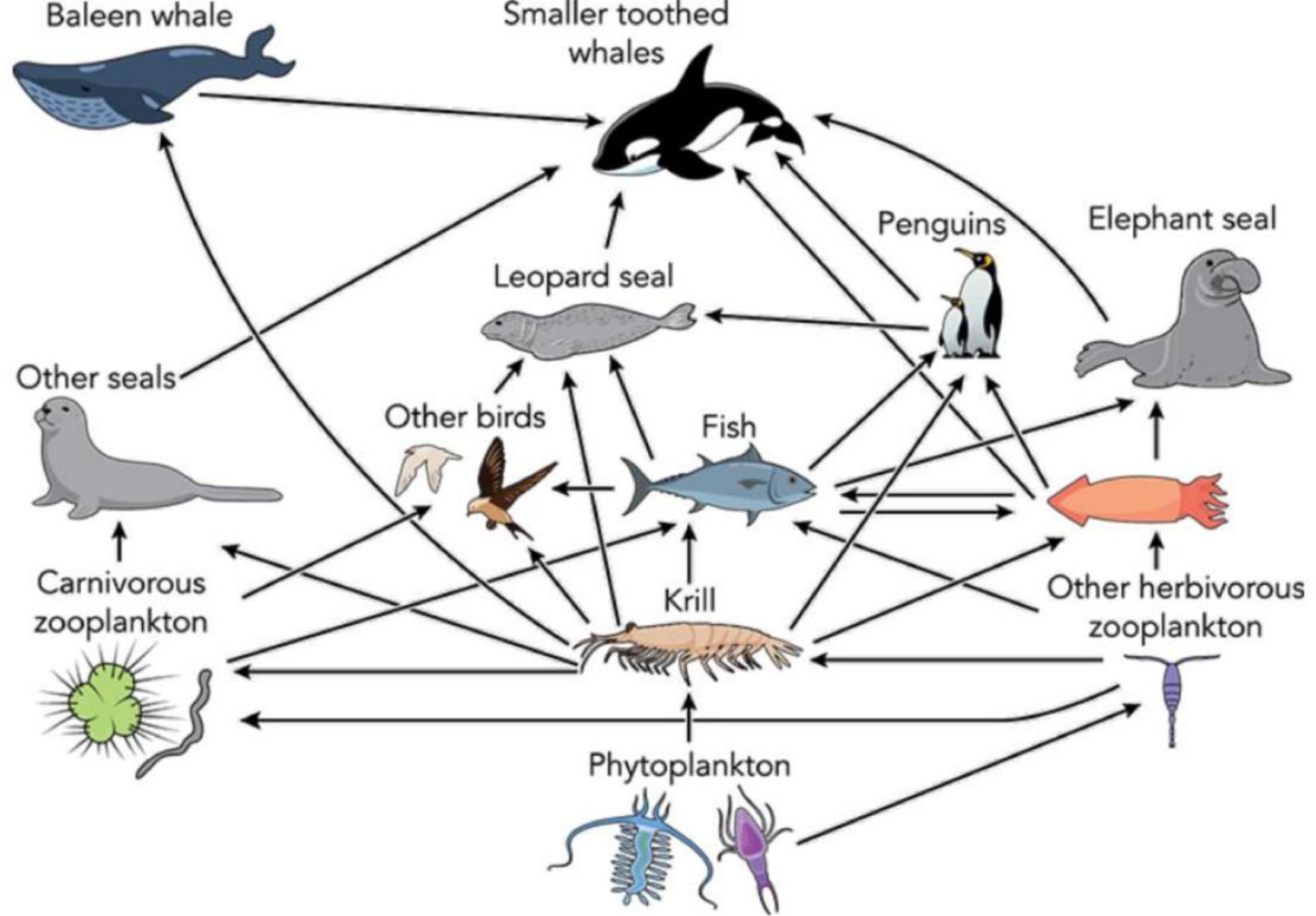
Antarctica offers a wide variety of resources and opportunities that humans use. For example:

- Food sources: the oceans around Antarctica are rich in marine life and support a diverse food web of organisms that humans harvest for food products (like whales, seals, and krill).
- Fresh water sources: approximately 60% of the Earth's total fresh water, an increasingly valuable resource, is stored in the Antarctic ice sheet. The increasing global population means water as a resource is coming under greater pressure, making Antarctica a potential source.
- Potential mineral sources: coal, oil, and other minerals have been found in Antarctica (though their mining is banned).
- Scientific exploration: the region offers unique opportunities for scientific research. Using ice-core drilling and sampling, researchers have gathered data about the Earth's climate from up to 800 000 years ago. This record helps us learn about human-driven climate change, by comparing modern greenhouse gas concentrations with the gas trapped in ancient ice.
- Tourism: Antarctica is increasingly becoming a destination for adventure travel. Although it is expensive, it is a desirable region for many tourists.

There are many interrelated threats to the Antarctic ecosystems. As with any resources, when they are over-utilised or over-harvested, the ecosystem comes under threat of damage.

Table 3.2: Threats and impacts to the Antarctic ecosystem

Threat	Impact
Overfishing	Excessive fishing for target species such as whales, seals, Antarctic cod, and krill causes these species populations to crash and changes food webs (Source A).
Resource exploitation	The mining of resources such as oil and coal would cause harm to the Antarctic ecosystems through both habitat destruction and the high risk of pollution. Commercial mining in Antarctica is currently prohibited.
Change to food webs	Overfishing of a specific fish species causes changes to food webs. A reduction of a specific species results in their prey increasing in numbers, while their predators increase their consumption of other prey species. This shifts the balance of biodiversity in the region.
Decreased biodiversity	The uncontrolled removal of primary consumers (such as krill) and apex predators (such as whales) results in decreased biodiversity, as the organisms that feed on them die out. The removal of primary consumers undermines the ecosystem's ability to function effectively and adjust to change.
Pollution	Plastics, micro-plastics, and oil spills all cause long-term harm to ecosystems. They result in the death of different organisms, reducing biodiversity.
Ecosystem damage	Mining (currently banned) and the construction of infrastructure (necessary to support human activities) causes damage to a fragile ecosystem that takes a long time to recover from disturbances.
Climate change and ozone layer depletion	Emissions of greenhouse gases and chlorofluorocarbons (CFCs) have the two-fold effect of increasing both temperature and incoming ultraviolet light in the Antarctic region. Both factors cause harmful changes to the ecosystem, such as melting ice and negatively affecting photosynthesising organisms.
Tourism	Tourism brings disturbances to Antarctica. For example, cruise ships increase the risk of sewage, oil spills, and other waste products like plastic causing pollution. Humans also cause noise pollution and disturb wildlife. This can change animals' behaviour patterns and interrupt the breeding or feeding of species. Visitors also bring in non-native species unintentionally, which can cause changes in local food webs.



Source A: An example of an Antarctic food web with the arrows showing the direction of flow of energy between prey and predator

► Image Description

TIP

When discussing changes in a food web, take note of which organism eats or is eaten by another organism. An increase or decrease in prey or predator will change the structure of the food web.

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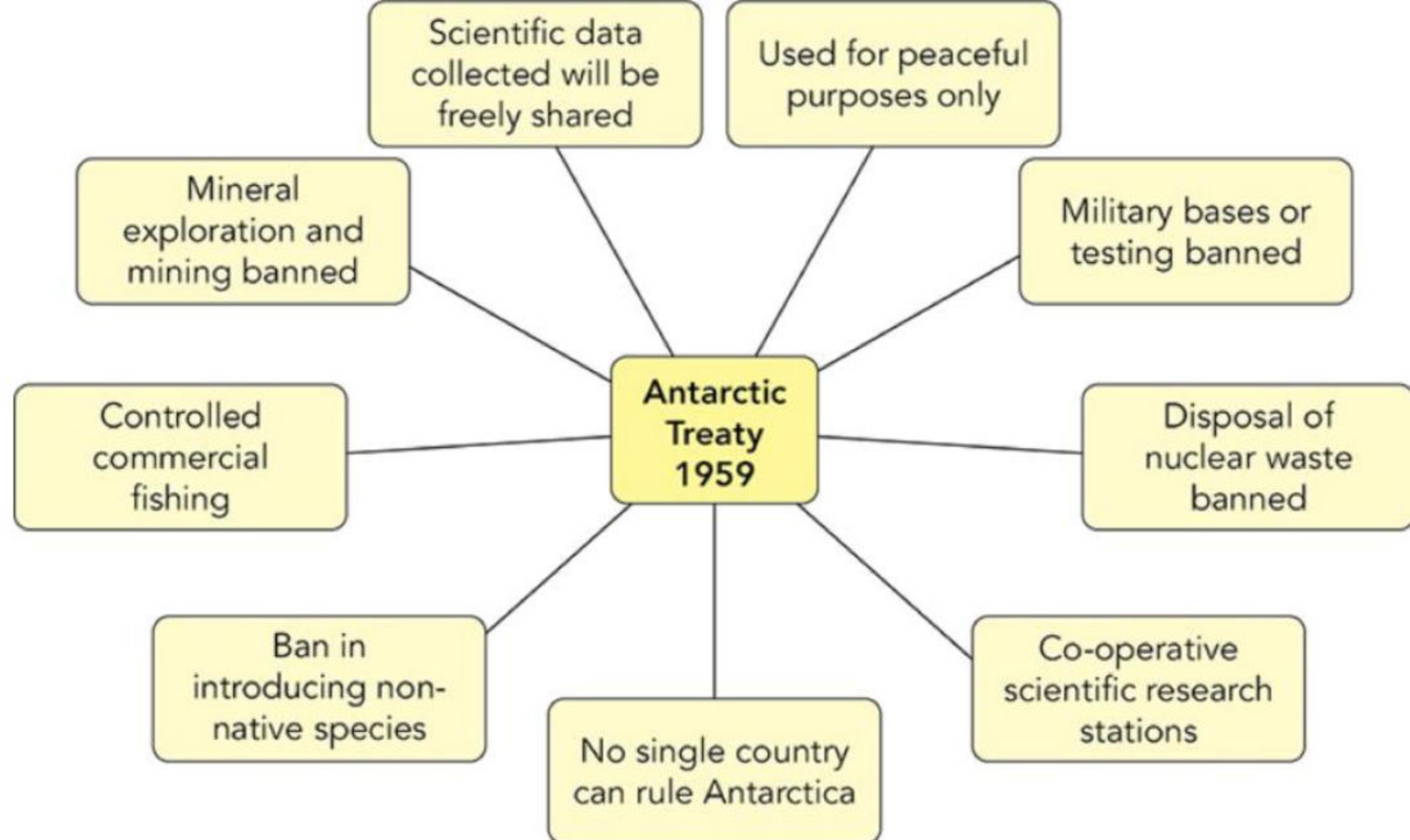
TASK 1: Study Source A

Primary producers (photosynthesising phytoplankton) form the base of the food web in Source A. Primary consumers feed on primary producers (krill, zooplankton). In turn, secondary consumers and apex predators feed on those organisms.

- Explain why the smallest organism in the food web, phytoplankton, is so important to the food web.
- Describe how the introduction of a non-native species that eats penguin eggs would impact the food web.
- Identify the organism that consumes the most diverse diet in the food web.
- With a partner, discuss the food web. Are there any organisms that have a diet that surprises you?

Sustainable management of the Antarctic ecosystem

Due to the pressures of fishing of the southern oceans, which started in the mid-1700s, the need for measures to ensure the sustainability of Antarctica was identified as early as the 1930s. There are a number of strategies and techniques in place to sustainably manage the Antarctic ecosystem, and these vary in their scope. A strategy is a long-term plan that has overarching goals (such as the Antarctic Treaty and other international agreements), while a technique focuses on making changes that will get immediate results (for example, establishing protected areas or training people visiting the area how to behave to minimise disruption to wildlife).



Source B: Some of the main rules found within the Antarctic Treaty (1959)

The Antarctic Treaty (1959) is an international agreement that established a strategy to protect Antarctica (see Source B). In addition to the Antarctic treaty, there are several other protocols and international agreements that protect wildlife in Antarctica:

- The International Whaling Commission (IWC) imposed a quota system in the 1970s on all parties hunting whales. By 1982, a limited timeframe ban was imposed on commercial whaling. However, by 1994, an indefinite ban was put on all whaling in waters south of Australia.
- Two large Marine Protected Areas (MPAs) have been established in the Antarctic to protect biodiversity through protecting those areas from commercial activities.
- The International Maritime organisation implemented a Polar Code in 2017 that limits the operation of very large cruise ships in Antarctica.
- The International Association of Antarctic Tour Operators (IAATO) is working on putting laws in place to manage tourism to the region to limit harm caused by human activity.
- The Montreal Protocol (1991) requires polluters or past users of Antarctica to clean up previously used work sites and waste dumps (for example, nuclear waste from military activities).

The success of the strategies and techniques used to protect Antarctica depends on international cooperation, and the ongoing development of the treaties to meet new threats as they arise. There are a number of challenges to managing the threats to Antarctica:

Tourism

The rapid increase in tourism in recent years has resulted in increased pressure on the ecosystem. In order to control this problem, tourism in Antarctica is being reviewed by various stakeholders. For example, there are discussions of the possibility of making anybody running a tour to Antarctica have to become a member of IAATO. This would help to manage both the numbers and behaviour of tour operators and tourists visiting the region.

Size of the ecosystem

Another challenge faced by parties trying to manage the environmental concerns of the region is the sheer size of the area. For example, it is difficult to ‘police’ the fishing that is taking place in the Southern Ocean, so ensuring that all parties are meeting international agreements is challenging.

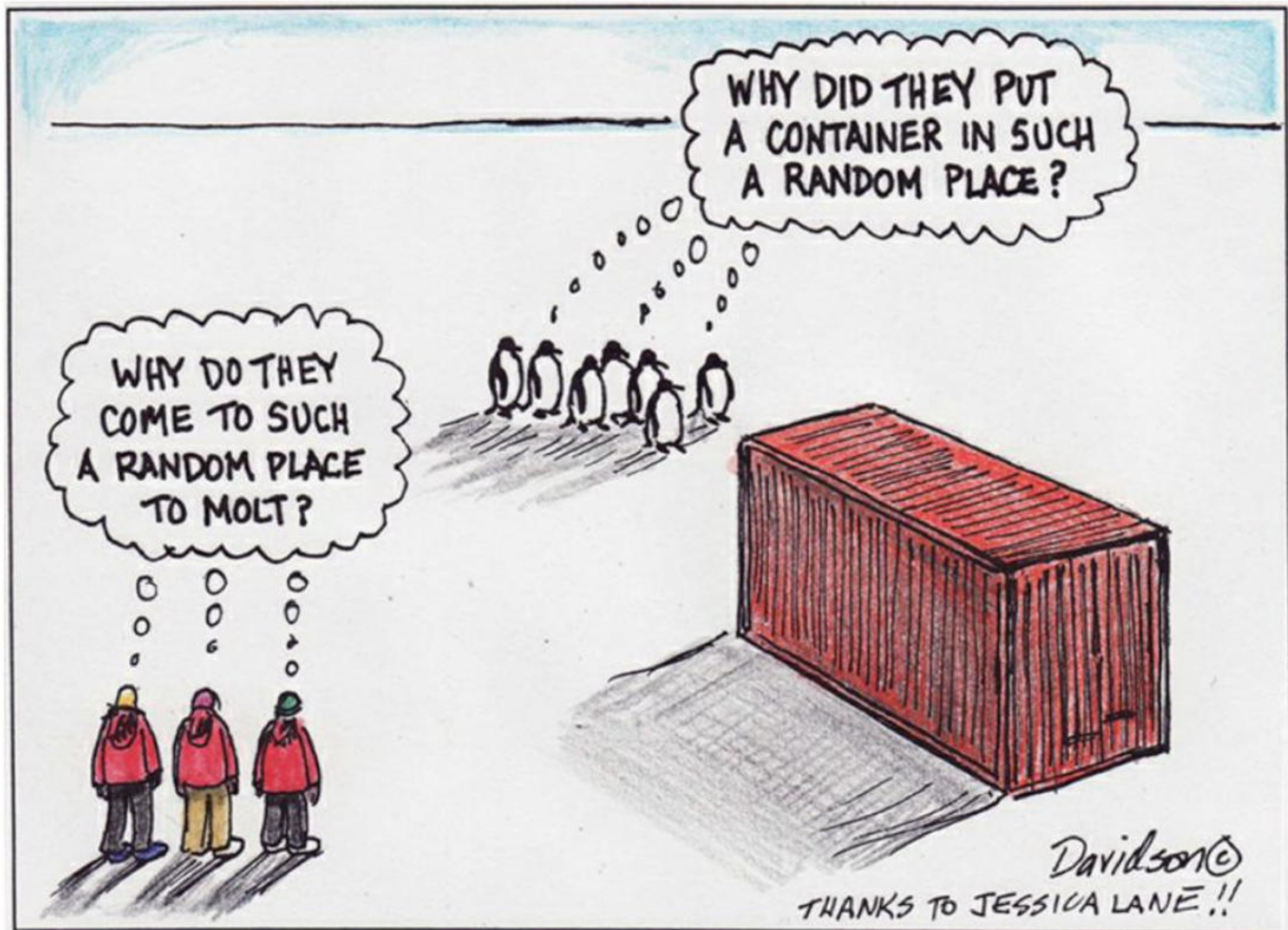
Global issues

The increase in greenhouse gases due to the burning of fossil fuels is a global concern that has a serious impact on Antarctica. The problem does not originate in the Antarctic region, though. The pollutants that are causing harm from thousands of km away. Reducing this threat is one that needs to be carried out by all countries around the globe. This is difficult to achieve and is therefore a limitation in managing the threats to the region.

There are, however, a number of success stories on the management of this ecosystem. For example, the international agreement that prevents the mining of resources in Antarctica, the closely managed research stations of different countries, and the fact that the region may not be governed by a single government have helped to reduce the impacts of humans on the continent. In addition, scientists based in Antarctica carry out environmental impact surveys which look at how changes caused by human activities are impacting the environment and causing changes in the region. These may look at how non-native species, pollution, or climate change are causing shifts in the ecosystems, and what action can be taken to try to minimise those changes in the future.

TASK 2: Study Source B

- a** The Antarctic Treaty of 1959 put in place a rule that no single country can own or rule Antarctica. In small groups, discuss this. Explain in a short, written summary why you think this is important.
- b** Explain why countries would want to test military equipment in Antarctica.
- c** Explain why you think countries should have cooperative scientific research and share data collected in Antarctica.
- d** Draw a spider diagram with sustainability at the centre and add how each of the international treaties (IWC, MPAs, Polar Code, IAATO and Montreal Protocol) all contribute towards sustainability.

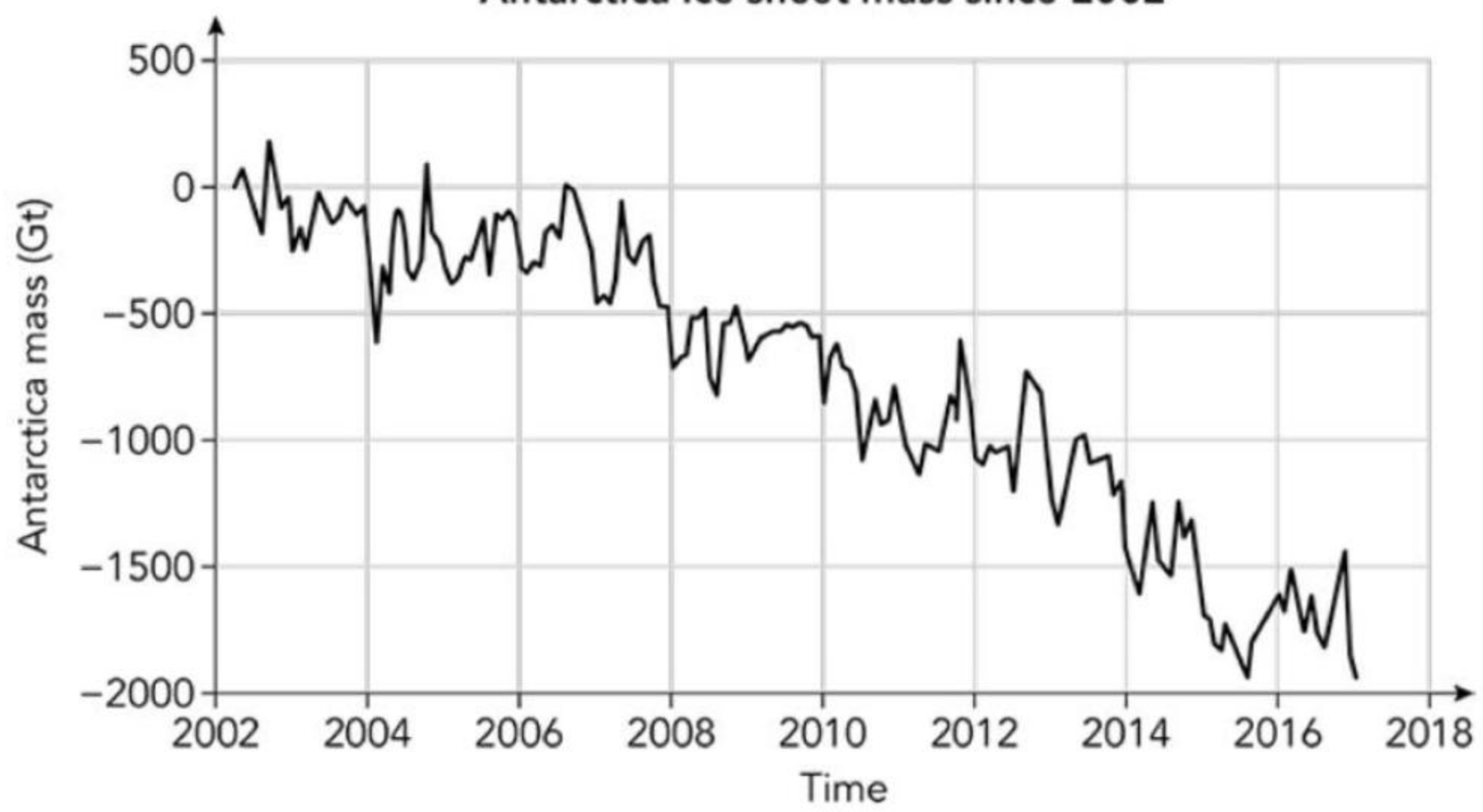


Source C: A cartoon showing difference of opinion on the location of a container

TASK 3: Study Source C

In small groups, discuss the message that the cartoon in Source C is trying to deliver. Then write a 100 to 150-word summary of your interpretation of the cartoon's message. Use knowledge that you have gained throughout this topic to support your viewpoint.

Antarctica ice sheet mass since 2002



Source **D**: Data collected by scientists on the decrease in Antarctic ice mass since 2002

TASK 4: Study Source **D**

- a **Describe the trend** in ice sheet mass between 2002 and 2017.
- b Explain why the ice sheets are thinning in Antarctica.